Computational Neuroscience

Introduction and Basic Neurobiology

Course Introduction

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1. What is Computational Neuroscience?

> "The goal of computational neuroscience is to explain in computational terms how brains generate behaviours."
>
> — Terry Sejnowski, one of the founders of the field

Computational neuroscience provides tools and methods for **characterising what** nervous systems do, **determining how** they function, and **understanding why** they operate in particular ways (Dayan & Abbott)

Those three verbs are not a rhetorical flourish — they map exactly onto the three classes of model, and this document works through all three on a single example: the receptive fields of the visual system.

Descriptive Models of the Brain *the **what***

How do neurons respond to external stimuli, and how do we describe this quantitatively with a **neural encoding model**?
How can we extract information from neurons — **decoding**?

Mechanistic Models of Brain Cells and Networks *the **how***

How can we **simulate** the behaviour of a single neuron on a computer?
How do we simulate a **network** of neurons?

Interpretive (or Normative) Models of the Brain *the **why***

Why do brain circuits operate the way they do?
What are the **computational principles** underlying their operation?

In one line each. **Descriptive** — what the system does. **Mechanistic** — how it does it. **Interpretive** — why it does it that way.

2. An Example: Models of Receptive Fields

Hubel and Wiesel, c. 1965 — recording from an intact cat brain

To understand the visual system of the cat, Hubel and Wiesel implanted **recording electrodes** into the visual area of the brain and recorded electrical signals from individual cells. Those signals are the cell's output: brief digital pulses called **action potentials**, or **spikes**.

[Lecture slide omitted from the public copy.] The experimental setup. A recording electrode in the visual area of the cat's brain picks up the electrical signal while a stimulus is presented on a screen. Right: the oscilloscope trace as an oriented bar is swept across the visual field.

They then presented different visual stimuli and watched which ones drove the cell. The result was strikingly specific:

- the cell fired vigorously for a **bar of light oriented at roughly** 45°;
- it responded weakly, or not at all, to **broad-field illumination**;
- it also responded to the **edge** of a stimulus presented at that same orientation.

[Lecture slide omitted from the public copy.] Same bar, three orientations. The horizontal bar (top) barely changes the firing rate. At 45° (middle) the cell fires a dense burst while the light is on. At the opposite oblique (bottom) the response is intermediate. Each vertical tick is one spike.

So the cell is not reporting "how much light is present". It is reporting something far more specific about the structure of the image.

Receptive field

The specific properties of a sensory stimulus that generate a strong response from a cell.

Examples:

- a **spot of light** switched on at a particular location on the retina;
- a **bar of light** at a particular orientation *and* location on the retina.

3. Descriptive Model: *What* Receptive Fields Look Like

Starting at the retina

The **retina** is the tissue at the back of the eye, onto which an inverted image of the world is projected. **Retinal ganglion cells** are the output neurons of the retina: they convey information about that image onward, first to the **lateral geniculate nucleus (LGN)** and then to cortex.

[Lecture slide omitted from the public copy.] Retinal ganglion cells are the retina's output neurons. Right: a spot of light in the centre of the receptive field drives a burst of spikes (top), while the same spot in the surround silences the cell, which then fires on light-off (bottom).

Flashing small circular spots of light at different retinal locations reveals the structure of a ganglion cell's receptive field:

- a spot in the **centre** of its receptive field produces a robust burst of spikes;
- a spot in the **surrounding** annulus **suppresses** firing — the cell falls silent, and often fires a burst when the light is switched *off*.

This opponent arrangement is called a **centre–surround receptive field**.

Centre–surround receptive fields in the retina

An **on-centre, off-surround** cell increases its firing when light falls on the centre and decreases it when light falls on the surround. An **off-centre, on-surround** cell does the opposite: it is suppressed by light in the centre and excited by light in the surround.

[Lecture slide omitted from the public copy.] The two polarities. + marks a region where light increases firing, − a region where light suppresses it. Shading marks the suppressive region in each case.

Looking ahead (the *why*). A filter that is excited by a bright centre and suppressed by a bright surround responds weakly to uniform illumination and strongly to *local contrast*. In other words, the retina is not transmitting a picture — it is transmitting where the image **changes**. That is an interpretive claim, and §5 picks it up.

From retina to cortex

Retina ⟶ Lateral geniculate nucleus (LGN) ⟶ Primary visual cortex (V1)

[Lecture slide omitted from the public copy.] The early visual pathway: retina → lateral geniculate nucleus → primary visual cortex.

Orientation preference in V1

Neurons in V1 respond best to a **bright bar on a dark background at a particular orientation** — for the cell Hubel and Wiesel recorded, about 45°. The same bar at other orientations produces markedly fewer spikes.

[Lecture slide omitted from the public copy.] The orientation tuning of the V1 cell (left) summarised as an oriented receptive field (right): an elongated excitatory band (+) flanked by inhibitory regions (−), aligned with the preferred orientation. Bottom: other oriented receptive fields found in V1 — differing in preferred angle, in the number of subregions, and in polarity.

Such a cell is said to have an **oriented receptive field**: an elongated region of the visual field with an excitatory band flanked by inhibitory regions. This is the **descriptive model** — it says *what* the cell does, without yet saying how the circuit achieves it or why it should.

V1 cells differ in their preferred angle, in how many excitatory and inhibitory subregions they have, and in polarity. Measuring these receptive fields quantitatively — rather than by eye — is done with **reverse correlation**, which the course returns to later.

4. Mechanistic Model: *How* They Are Built

We now have circular **centre–surround** fields in the retina and the LGN, and elongated **oriented** fields in V1. The obvious next question is how the second is built out of the first, using the actual circuitry of the cortex.

The clue is in the anatomy

LGN receptive fields are still centre–surround — much like the retina's — while V1 fields are oriented. So the transformation happens between LGN and V1. The anatomical fact that solves it: **many LGN cells converge onto a single V1 cell**. One V1 neuron is reading out a large population of centre–surround inputs at once.

[Lecture slide omitted from the public copy.] Left: the pathway from eye to LGN to V1. Right: LGN receptive fields are centre–surround, V1 receptive fields are oriented — and many LGN cells converge onto one V1 cell.

The answer

It is almost obvious once stated. To build an elongated receptive field, take LGN cells whose (circular) receptive fields are **displaced along a line**, and let them all feed the same V1 cell.

That V1 cell now responds best to a bright bar lying along that line — it behaves exactly like the cell in the Hubel and Wiesel recording.

[Lecture slide omitted from the public copy.] Centre–surround LGN receptive fields, aligned along the preferred orientation and converging on one V1 cell, produce an oriented V1 receptive field. Proposed by Hubel and Wiesel in the 1960s.

Why the model is still controversial

The model is purely **feedforward**: LGN in, V1 out. But the anatomy of V1 shows a great many **recurrent connections** between V1 cells. Every V1 neuron receives input from its neighbours in V1 as well as from the LGN, and those recurrent inputs demonstrably shape its response.

Worth holding onto. The Hubel–Wiesel construction accounts for the orientation tuning without invoking recurrence at all — which is what makes it elegant, and also what makes it incomplete. Feedforward *and* recurrent networks are treated properly later in the course.

5. Interpretive Model: *Why* They Are Shaped That Way

The question

Why should receptive fields be oriented, localised, and selective for bright or dark bars? This is the question an engineer would ask: what are the **computational advantages** of building a visual system this way?

The efficient coding hypothesis

Suppose the goal — reached through evolution — is to represent images as **faithfully** and as **efficiently** as possible, using the neurons available. Each neuron i has a receptive field RF_i and a response r_i.

An image can then be **reconstructed** as a weighted sum of receptive fields, where the weights are the neural responses:

$$\hat{I} = \sum_i \mathrm{RF}_i \, r_i$$

[slide omitted]

Add two receptive fields together and you get a new pattern — bright where their + regions overlap, dark where the − regions do. With enough receptive fields, any image patch can be approximated this way.

Turning it into an optimisation

The hypothesis becomes concrete as a search for the RF_i that satisfy two demands at once:

- **Faithful** — minimise the total squared pixelwise error between an image I and its reconstruction $\hat{I}$;
- **Efficient** — keep the responses r_i as **independent** as possible. We do not want every neuron firing at once; that would be a redundant code.

Running it on natural images

Start from **random** receptive fields, then run an efficient coding algorithm on small patches sampled from **natural images** — on the assumption that the brain evolved to encode the kind of scenes an animal actually looks at.

[Lecture slide omitted from the public copy.] Natural image patches. The algorithm samples small windows at random locations; the red square indicates the size of one receptive field relative to the image.

Three algorithms do this, differing in how they enforce efficiency:

- **Sparse coding** — Olshausen & Field, 1996
- **Independent component analysis (ICA)** — Bell & Sejnowski, 1997
- **Predictive coding** — Rao & Ballard, 1999

The result

The receptive fields that emerge are **oriented**, **localised** to particular positions, and have adjacent bright and dark regions — the same + and − structure seen in V1, spanning the range of orientations present in natural scenes. Nobody put that structure in; it fell out of the optimisation.

[Lecture slide omitted from the public copy.] Top: receptive fields learned from natural images by an efficient coding algorithm (white = +, dark = −). Bottom: receptive fields measured in V1. The learned fields recover the oriented, localised, biphasic structure of the real ones.

Conclusion. The brain may be trying to find faithful and efficient representations of an animal's natural environment. The same principle applied to *sounds* predicts auditory cortex representations too — so this is not a fact about vision, but a candidate principle for sensory coding in general.

6. The Three Models, Side by Side

Descriptive — V1 cells respond best to a bar at a particular orientation. *(what)*

Mechanistic — because centre–surround LGN cells, aligned along that orientation, converge onto them. *(how)*

Interpretive — because oriented, localised filters are an efficient code for natural images. *(why)*

One further thread runs through all three: at every stage the description takes the same form, **response** = f(**stimulus**). Centre–surround fields in the retina detect local contrast; oriented fields in V1 detect edges at specific angles; each stage extracts more structure than the last.

What none of the three has said yet is what a neuron physically *is*. That is where the next lecture starts.

7. Enter the Neuron

The first three sections treated the neuron as an abstraction — something that turns a stimulus into a firing rate. This part opens the box and asks what the physical cell actually is.

A cortical neuron is tiny: the cell body is about **25 microns** across, a micron being a millionth of a metre. From that cell body run many branches, the **dendrites**, and one long slender fibre, the **axon**, which carries the cell's output away.

[Lecture slide omitted from the public copy.] Left: a single neuron in visual cortex, filled green against the surrounding tissue; the cell body is roughly 25 μm across. Right: where that cell sits — cerebral cortex, with cerebellum and spinal cord below.

The neuronal zoo

There is no single shape of neuron. Ramón y Cajal's drawings from around 1900 show **pyramidal cells** in cortex (named for their triangular cell bodies, and for the pyramidal tract their axons form), **Purkinje cells** in cerebellum with their extraordinary flat dendritic trees, and yet other branching patterns in the optic tectum. Morphology varies systematically with brain region — and, within a region, with depth.

[Lecture slide omitted from the public copy.] Three regions, three morphologies. Left: visual cortex. Middle: cerebellum, with two Purkinje cells and their densely branched dendrites. Right: optic tectum, where branching pattern varies with depth.

Those drawings settled a genuine controversy. One camp — the **reticular hypothesis** — held that the brain was one continuous network. The other held that it was made of separate cells. Cajal's observations were clear enough to decide it.

The neuron doctrine

- The neuron is the fundamental **structural and functional unit** of the brain.
- Neurons are **discrete cells**, not continuous with one another.
- Information flows from the **dendrites**, via the cell body, to the **axon**.

There are exceptions to each clause, but they hold for the majority of neurons and brain regions.

The idealized neuron

Strip that down and you get the model the rest of the course actually uses. Inputs arrive from the axons of other neurons. Each input produces a small positive deflection in the receiving cell, an **EPSP** — excitatory post-synaptic potential. The cell **sums** these over time and over location along its dendrites, and if the sum reaches a **threshold**, the cell fires an **action potential**, or **spike**, which travels out along the axon.

[Lecture slide omitted from the public copy.] The idealized neuron. Dendrites and cell body collect inputs; the axon carries the output. Bottom left: individual EPSPs. Bottom right: the membrane sits at a resting potential of about $-70\,mV$ and produces a spike when the summed input crosses threshold.

8. What a Neuron Actually Is

A leaky bag of charged liquid

Unflattering, but accurate. It is a **bag** because the contents are enclosed in a **cell membrane** — a **lipid bilayer**, which is to say a double sheet of fat. That bilayer is **impermeable** to charged

ions such as Na^+, Cl^- and K^+, which are dissolved both outside and inside the cell. It is **leaky** because embedded in the membrane are **ionic channels** that let specific ions cross. Everything interesting happens at those channels.

The electrical personality of a neuron

Each neuron holds a **potential difference** across its membrane: the inside sits at roughly **−70 mV** relative to the outside. This is the **resting membrane potential** — the state when no spike is happening.

[Lecture slide omitted from the public copy.] The resting membrane. Na^+ and Cl^- are more concentrated outside, K^+ and organic anions $[A^-]$ inside; the interior sits at $-70\,mV$ relative to the exterior.

The asymmetry is maintained by an **ionic pump** that actively expels Na^+ and draws K^+ in. Active means it costs energy — a substantial share of the brain's energy budget goes on simply holding this difference in place.

9. Ionic Channels: The Gatekeepers

Ionic channels are **proteins** embedded in the membrane, and they are **selective**: a channel may pass Na^+ but not K^+ or Cl^-. They are also **gated** — they open and close depending on local conditions. Three kinds:

- **Voltage-gated** — the probability of being open depends on the local membrane voltage.
- **Chemically-gated** — a chemical binds to the channel and opens it. *Synapses are chemically-gated channels*, which is where the next lecture starts.
- **Mechanically-gated** — sensitive to pressure or stretch.

How gated channels let neurons signal

Put the two channel types together and the whole chain follows:

- Spikes arrive from another neuron at a **synapse** and open **chemically-gated** channels there.
- Na^+ diffuses inward — it is more concentrated outside — and being positive, it raises the local membrane potential.
- That voltage change opens or closes **voltage-gated** channels nearby, producing **depolarization** (voltage becomes more positive) or, if K^+ channels open and positive charge leaves, **hyperpolarization**.
- A large enough depolarization crosses threshold and a spike is produced.

[Lecture slide omitted from the public copy.] Inputs arriving at synapses — the junctions between neurons. Each yellow terminal marks a synapse onto the large cell at the left.

10. The Action Potential

The spike's characteristic shape is entirely a consequence of voltage-gated channels opening and closing in sequence:

1. A strong depolarization opens **Na^+ channels**. Na^+ rushes in, which depolarizes further, which opens still more Na^+ channels — a **positive feedback loop** that drives the sharp rising edge, until those channels **inactivate**.

2. At roughly the same moment, **K^+ channels** open. K^+ — more concentrated inside — flows out, carrying positive charge with it and pulling the membrane back down. That is the falling edge. The K^+ channels then close, completing the spike.

[Lecture slide omitted from the public copy.] One action potential, annotated by what the channels are doing. Yellow: Na^+ channels opening. Green: more Na^+ channels opening (positive feedback). Cyan: Na^+ channels closing while K^+ channels open and restore the potential.

Why this matters for the models. Because the shape is fixed by channel dynamics, it is *stereotyped* — every spike from every neuron looks essentially the same. No information is carried in the shape of a spike. Whatever a neuron communicates must therefore be carried in *which* neuron fires and *when* — which is exactly why the descriptive models of §3 were written as functions of firing rate.

Propagation, and the myelination trick

Once generated at the start of the axon, the spike propagates: at each point along the axon, Na^+ channels open (rising edge), then close as K^+ channels open (falling edge), regenerating the spike a little further along.

Axons can do this faster. Glial cells called **oligodendrocytes** wrap the axon in insulating sheaths of **myelin**, leaving bare gaps called the **nodes of Ranvier**. Charge spreads passively through the cytoplasm under each sheath and the spike is regenerated only at the nodes, so the action potential effectively *hops* from node to node — **saltatory conduction**. The result is fast, lossless long-range signalling: an "active wire".

When it fails. In **multiple sclerosis**, an autoimmune response destroys myelin. The axons survive but lose the ability to conduct spikes quickly and reliably, which is why the symptoms are so varied — they depend on which pathways happen to be demyelinated.

[Lecture slide omitted from the public copy.] An oligodendrocyte wrapping myelin sheaths around an axon, leaving nodes of Ranvier exposed between them. Inset: the layered sheath in cross-section.

Summary

A neuron has a cell body, dendrites that collect input, and an axon that carries output. The output is a train of stereotyped electrical impulses — spikes — generated when summed input crosses threshold, produced by voltage-gated Na^+ and K^+ channels, and delivered at junctions called **synapses**.

What happens once a spike arrives at a synapse is the subject of the next lecture.

11. The Synapse

> The ionic mechanism of the action potential was turned into a mathematical model by **Hodgkin and Huxley** in 1952 — work that won them a Nobel Prize, and which the course returns to later. What follows first is what happens to a spike once it reaches the *end* of the axon.

A **synapse** is a connection — a junction — between two neurons. By convention the sending cell is the **presynaptic** neuron and the receiving cell the **postsynaptic** neuron. There are two kinds.

[Lecture slide omitted from the public copy.] Left: an electrical *synapse. Gap junctions — ion channels spanning both membranes at once — connect the interiors of neuron A and neuron B directly. Right: a*

chemical *synapse. An arriving spike triggers vesicles to release neurotransmitter into the cleft, where it binds receptors on the postsynaptic side.*

Electrical synapses

Gap junctions are ion channels that span the membranes of both cells, so the interiors are directly connected. Excitation on one side — say a higher sodium concentration following an action potential — lets ions migrate straight across, changing the membrane potential of the second cell. This behaves much like a wire between two components: fast and direct.

That speed is the point. Electrical synapses turn up where neurons need to be **synchronised** — made to fire together — and where a reflex must be as fast as possible, such as the escape reflex of the crayfish.

Chemical synapses

The chemical synapse takes a longer route. The presynaptic terminal holds **vesicles**: small bags of **neurotransmitter** molecules. When a spike arrives, the vesicles fuse with the membrane and release their contents into the **synaptic cleft**, the gap between the cells. The neurotransmitter binds receptors on the far side — and those receptors are exactly the **chemically-gated ion channels** of the last lecture. They open, ions flow according to their concentration gradients, and the postsynaptic membrane potential changes.

So the signal runs **electrical → chemical → electrical**.

Why go to the trouble? A gap junction would have been simpler, so why build the chemical detour? Because it makes the connection *adjustable*. The postsynaptic cell can change how much the same presynaptic spike affects it, simply by changing the **number or density of receptor channels** on its side. Remove some, and the identical spike from A now produces less excitation in B. That single degree of freedom is why chemical synapses are the candidate substrate for learning and memory.

How many, and of what sign

A single neuron carries up to **10,000 synapses** on its dendrites and cell body.

[Lecture slide omitted from the public copy.] One neuron and its synapses (Kennedy lab, Caltech). Each bright spot is a single synapse. The resemblance to a city seen at night from the air is hard to avoid.

Excitatory and inhibitory synapses

- An **excitatory** synapse tends to *increase* the postsynaptic membrane potential — it pushes neuron B towards firing.
- An **inhibitory** synapse tends to *decrease* it — it pushes B away from firing.

The excitatory case, step by step:

input spike → neurotransmitter release (e.g. **glutamate**) → binds ion-channel receptors → channels open → Na^+ influx → depolarization → **EPSP**

[Lecture slide omitted from the public copy.] An excitatory synapse. Vesicles release neurotransmitter into the cleft; it binds receptors on the postsynaptic membrane and opens them, admitting Na^+ and depolarizing the cell.

12. The Synapse Doctrine: How Brains Learn

The synapse doctrine

Synapses are the basis for **memory and learning** in the brain.

Taken literally this is a strong claim: every memory you have, and every skill — typing, dancing, driving, reading — is stored in the strengths of your synapses.

The mechanism is **synaptic plasticity**: synapses change their strength with use. The oldest and most influential rule is due to the Canadian psychologist **Donald Hebb**, in 1949.

[slide omitted]

Hebb's rule. *If neuron A repeatedly takes part in firing neuron B, then the synapse from A to B is strengthened.*

In the diagram, A fires a burst and B fires once (top). Because A took part in making B fire, the A→B synapse is strengthened — drawn as the larger red terminal. The next time A fires the same burst, B responds more strongly and sooner (bottom).

The mantra: **"neurons that fire together wire together"**.

What the rule buys you. It is a *prediction* mechanism. Suppose you are in a dark jungle and hear a growl — neuron A fires. An instant later you see a tiger — neuron B fires — and you run. Hebbian plasticity strengthens A→B. The next time you hear a growl, A alone is now enough to drive B, and you start running *before* the tiger appears. The rule converts a temporal correlation into an anticipatory response.

The experimental evidence: LTP and LTD

Hebb's rule was a proposal. What the laboratory actually observes are two long-lasting changes in synaptic strength, most prominently in the **hippocampus**. Both are measured the same way — by the size of the EPSP that a single input from A produces in B.

- **Long term potentiation (LTP)** — an experimentally observed **increase** in synaptic strength, lasting hours or days. Pair A with B repeatedly and the same single input from A comes to produce a visibly larger EPSP.
- **Long term depression (LTD)** — the mirror image: an experimentally observed **decrease** in synaptic strength, again lasting hours or days, seen as a shrinking EPSP for the same input.

[Lecture slide omitted from the public copy.] Left (LTP): repeated pairing makes the EPSP produced by one input from A grow over time. Right (LTD): the same measurement, running the other way. In both panels the input from A is unchanged — what changes is the synapse.

13. Spike-Timing Dependent Plasticity

There is a twist. Whether a synapse potentiates or depresses turns out to depend on the **relative timing** of the input and output spikes — not merely on whether both occurred.

- Input spike **before** output spike (A fires, then B fires): the EPSP grows — **LTP**.
- Input spike **after** output spike (B fires, then A fires): the EPSP shrinks — **LTD**.

[Lecture slide omitted from the public copy.] The same pairing experiment run in both orders. Left column: the input precedes the output, and the EPSP measured afterwards is larger than before. Right column: the

input follows the output, and the EPSP shrinks.

Sweeping the interval between the two spikes traces out a **plasticity window**. It is remarkably sharp: within roughly ±40 ms of coincidence, the *sign* of the change flips depending on which spike came first.

[Lecture slide omitted from the public copy.] The STDP window (Bi & Poo, 1998). $\Delta t > 0$ — input before output — gives potentiation; $\Delta t < 0$ — input after output — gives depression. The change in synaptic strength is largest for the shortest intervals and dies away within a few tens of milliseconds.

Why the asymmetry matters. The window is not symmetric, and that is the point: it rewards the case where A's firing could plausibly have *caused* B's, and punishes the reverse. A synapse obeying this rule is learning temporal order, not just correlation — which is what makes it useful for learning *sequences* and generating predictions. Neurons in several areas, cerebral cortex among them, are sensitive to timing at this resolution.

14. What We Do Not Know

We now know a good deal about ionic channels, about single neurons, and about synapses. What we know about how *networks* of neurons give rise to perception, behaviour and perhaps consciousness is a great deal less.

That gap is the stated motivation for the large-scale brain projects: the **Human Brain Project** in Europe, led by Henry Markram, aiming at a large-scale computer simulation of the brain, and the **BRAIN Initiative** in the US, aiming to record hundreds of thousands of neurons simultaneously.

Next: brain organisation, and information processing in networks of neurons.

15. Organization of the Nervous System

The nervous system divides in two: the **peripheral nervous system (PNS)** and the **central nervous system (CNS)**. The PNS divides again into the **somatic** and **autonomic** systems; the CNS is the **brain** and the **spinal cord**.

[Lecture slide omitted from the public copy.] The four-way split. Somatic (skin and skeletal muscle) and autonomic (heart, vessels, glands) make up the periphery; brain and spinal cord make up the centre.

The peripheral nervous system

First, a definition that will keep recurring: a **nerve** is nothing but a **bundle of axons** — a cable made of the output fibres of many neurons.

The **somatic** system is the nerves connecting to your voluntary skeletal muscles and your sensory receptors. Its fibres come in two directions:

- **Afferent** (incoming) fibres — axons carrying information *from the periphery to the CNS*;
- **Efferent** (outgoing) fibres — axons carrying information *from the CNS out to the periphery.*

[slide omitted]

Shaking a friend's hand uses both at once: efferent fibres carry the commands that move your arm and hand, and afferent fibres carry back the sensation from the receptors in the skin of your palm.

The **autonomic** system is the nerves connecting to the internal organs — heart, blood vessels, smooth muscle, glands. It largely operates **below the level of consciousness**, regulating heart rate, digestion and respiratory rate, and it is responsible for the fight-or-flight response.

16. The Central Nervous System

The spinal cord

Reach into the oven, touch the hot pan, and your hand is already withdrawn before you have said anything about it. That is a **reflex arc**, and it is implemented entirely in the **spinal cord**: sensory information reaches spinal neurons, which directly drive the motor neurons that pull the hand away. The brain is not in the loop.

[Lecture slide omitted from the public copy.] A reflex arc. The sensory neuron signals the spinal cord, a relay neuron passes it to a motor neuron, and the muscle contracts — a local loop that does not wait for the brain.

If that were all the cord did we would live lives made entirely of reflexes. It also carries traffic in both directions:

- **Descending motor control** signals from the brain activate spinal motor neurons in specific patterns. The brain can instruct the cord to *walk* and then get on with something else — talking on the phone, say.
- **Ascending sensory axons** carry information from muscles and skin back up to the brain. Trip over an unseen rock and this is the path that interrupts the phone call.

17. A Tour of the Brain

[Lecture slide omitted from the public copy.] Four stops on the same mid-sagittal section. Top left: the hindbrain (cerebellum, medulla oblongata, pons). Top right: midbrain and reticular formation. Bottom left: thalamus and hypothalamus. Bottom right: the cerebrum, whose outer sheet is the cerebral cortex.

The hindbrain

- **Medulla oblongata** — breathing, muscle tone, blood pressure.
- **Pons** — sleep and arousal; also the link to the cerebellum.
- **Cerebellum** — coordination and timing of voluntary movements, sense of equilibrium, and more recently implicated in language and attention.

Midbrain and reticular formation

The **midbrain** handles eye movements and visual and auditory reflexes — a loud bang behind you makes you turn towards it, and that orienting reflex is midbrain work. The **reticular formation** modulates muscle reflexes, breathing and pain perception, and regulates sleep, wakefulness and arousal.

Thalamus and hypothalamus

The **thalamus** is a collection of **nuclei** — anatomically defined clusters of neurons. It is traditionally described as the **relay station** for sensory information on its way to cortex, and it also regulates sleep and wakefulness.

One sense is exempt. *Smell* does not go through the thalamus at all — olfactory input reaches cortex directly.

Immediately below it, the **hypothalamus** regulates basic needs — fighting, fleeing, feeding and mating.

The cerebrum

The **cerebrum** comprises the **cerebral cortex** together with the **basal ganglia**, **hippocampus** and **amygdala**. Between them they cover perception and motor control, cognitive function, emotion, memory and learning.

18. The Cerebral Cortex: A Layered Sheet

The cortex is the convoluted outer surface of the cerebrum, and it is only about **an eighth of an inch thick** — Christof Koch's image is of a 14-inch pizza folded up to fit inside the skull. The numbers in that sheet:

- roughly **30 billion neurons**;
- each making about **10,000 synapses**;
- hence on the order of **300 trillion connections**.

[Lecture slide omitted from the public copy.] Zooming in: the cerebrum, a stained cortical section, and the layered arrangement of neurons within the sheet.

Six layers, and a suspicious regularity

The sheet has **six layers** of neurons, and — this is the interesting part — it looks **relatively uniform across cortical areas**. Handed a piece of cortical tissue from one area and a piece from another, you would be hard pressed to tell them apart. The wiring is regular too:

- input from the thalamus terminates in the middle layers, **layer 4**;
- output back to the thalamus leaves from **layer 6**;
- output to other subcortical regions leaves from **layer 5**;
- output to "higher" cortical areas — those handling more abstract information — leaves from the superficial layers, **2 and 3**;
- feedback *from* those higher areas terminates in **layer 1**.

[Lecture slide omitted from the public copy.] The canonical cortical circuit: where input arrives and where each class of output leaves, by layer.

The open question. If a single circuit design is repeated across cortex — vision, hearing, touch, language — then perhaps a single **computational principle** is being applied to all of them. What that principle is, nobody knows. It is one of the field's genuinely open problems, and an unusually well-posed one.

19. Neural versus Digital Computing

	Brain	Silicon chip
Device count	10^{11} neurons, each with $\sim 10^4$ connections	10^{10} transistors, sparsely connected
Device speed	$\sim 100\,\mu$s temporal resolution	approaching a 100 ps clock (10 GHz)
Paradigm	massively parallel; *adaptive* connectivity	sequential processing; *fixed* connectivity
Good at	ill-posed problems — speech, vision	mathematics and symbol processing

Digital circuits beat biology on raw speed by six orders of magnitude, and are comparable on device count. The brain nevertheless wins at fast object recognition and speech. The difference is not the components but the **paradigm**: massive parallelism, and connections that change their own strength — the LTP and LTD of §12.

20. Where This Leaves Us

How all these regions interact to produce cognition and behaviour is not known. The honest answer is that we are inching towards it, on the back of electrophysiological, optical, molecular, functional-imaging, psychophysical, anatomical, connectomic and lesion studies.

What the structure of the brain suggests.

Information storage — in the physical and chemical structure of neurons and synapses.

Information transmission — electrical *and* chemical signalling: spikes along axons, neurotransmitters across synapses.

Primary computing element — the neuron.

Computational basis — currently unknown.

That last line is the whole reason for the three-model framework this document opened with. The course attacks the unknown from all three directions at once: **descriptive** models of what neurons do, **mechanistic** models of how they do it, and **interpretive** models of why.

Next week: *descriptive models of neural encoding and decoding*, with Adrienne Fairhall.